



# *VLSI Testing* *積體電路測試*

## **Testability Measure**

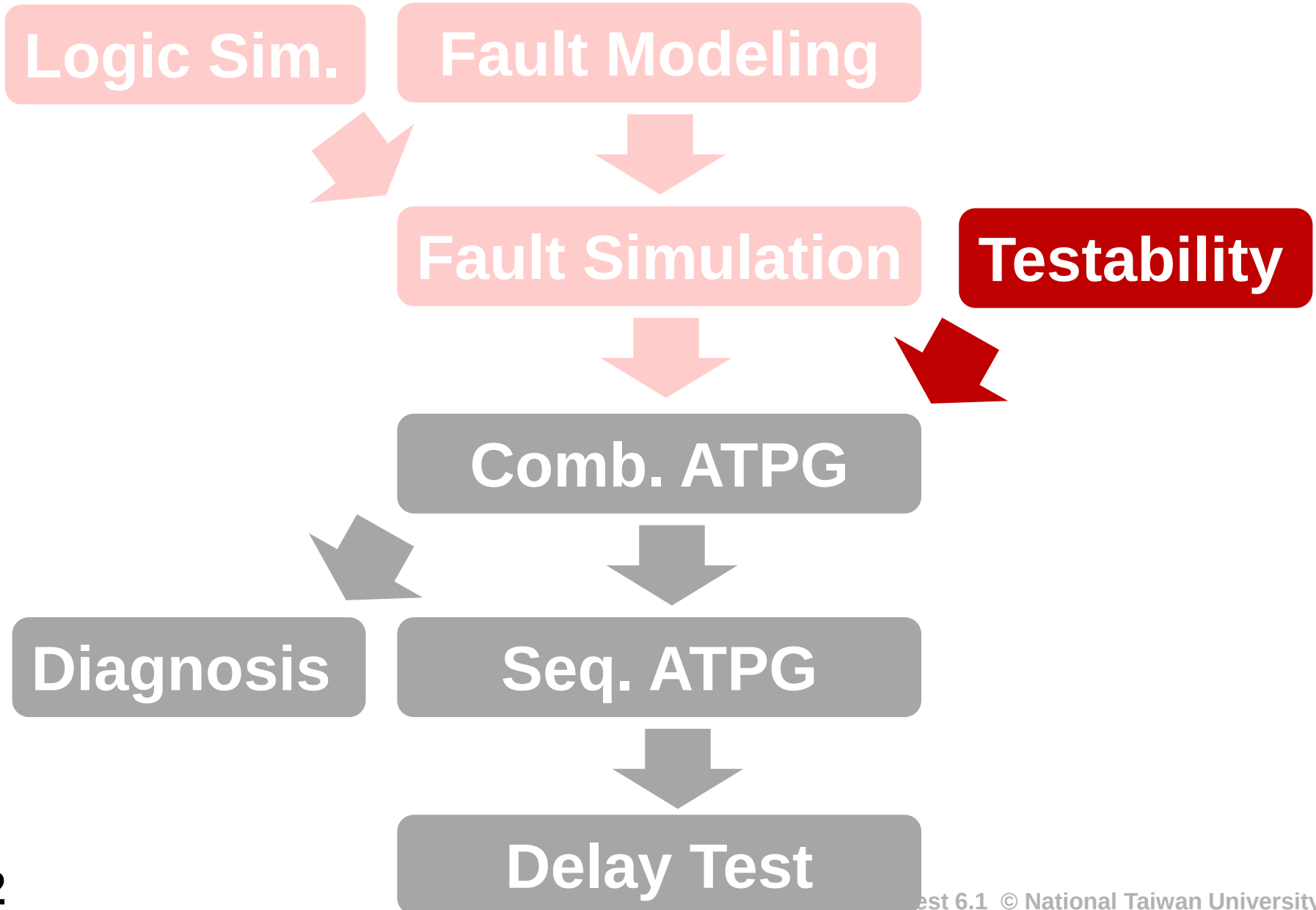
**Professor James Chien-Mo Li 李建模**

**Lab. of Dependable Systems**

**Graduate Institute of Electronics Engineering  
National Taiwan University**

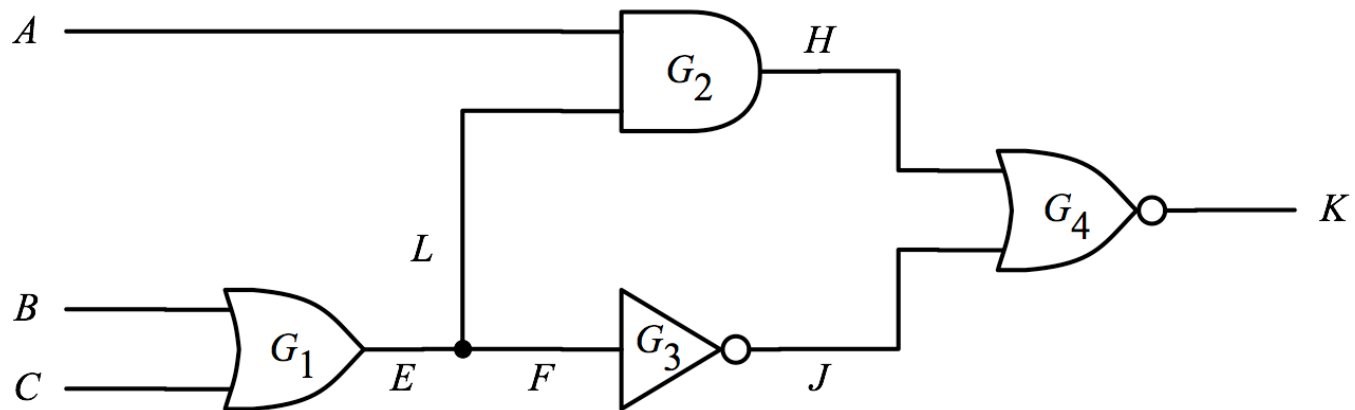
\* Some pictures are courtesy of Prof. Jiun-Lang Huang, NTU

# Course Roadmap (EDA Topics)



# Motivating Problem

- You report fault coverage of test set to your manager. Your manager is not very happy about FC number. He asked you: which faults are so difficult to detect?



# Why Am I Learning This?

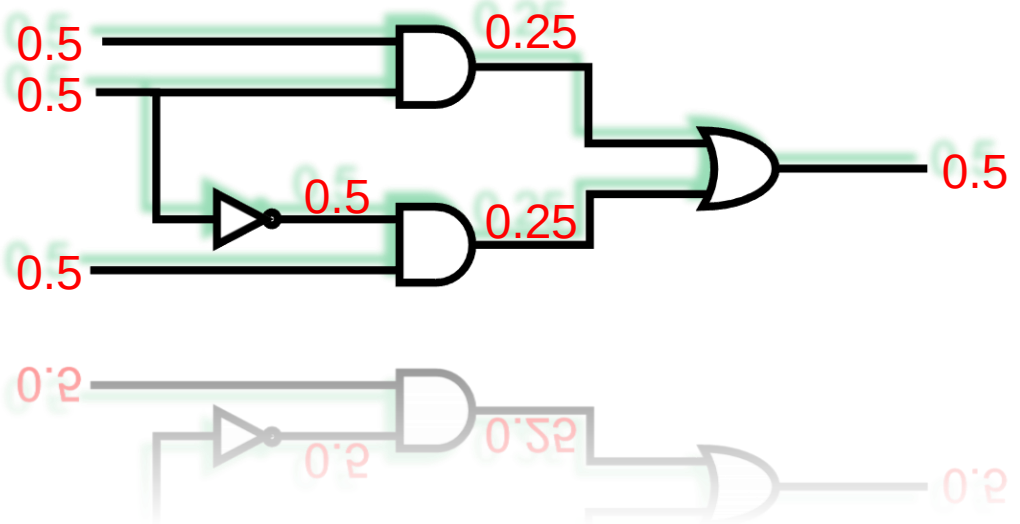
- Testability measure helps to
  - ♦ Make smart decision in ATPG
  - ♦ More testable designs

*“Measurement is the first step that leads to control and eventually to improvement. ”*

*(H. James Harrington)*

# Testability Measure

- Introduction
- SCOAP (1979)
- COP (1984)
- High-level Testability Measures
- Conclusion



# Testability Measures

- **What is testability measure?**
  - ♦ Metric to measure degree of difficulty to test a circuit
- **Two important components:**
  - ♦ **Controllability**
    - \* degree of difficulty to control a logic signal to 0 or 1
  - ♦ **Observability**
    - \* degree of difficulty to observe the logic value of a signal

**Testability Measures Controllability and Observability**

# Testability Analysis

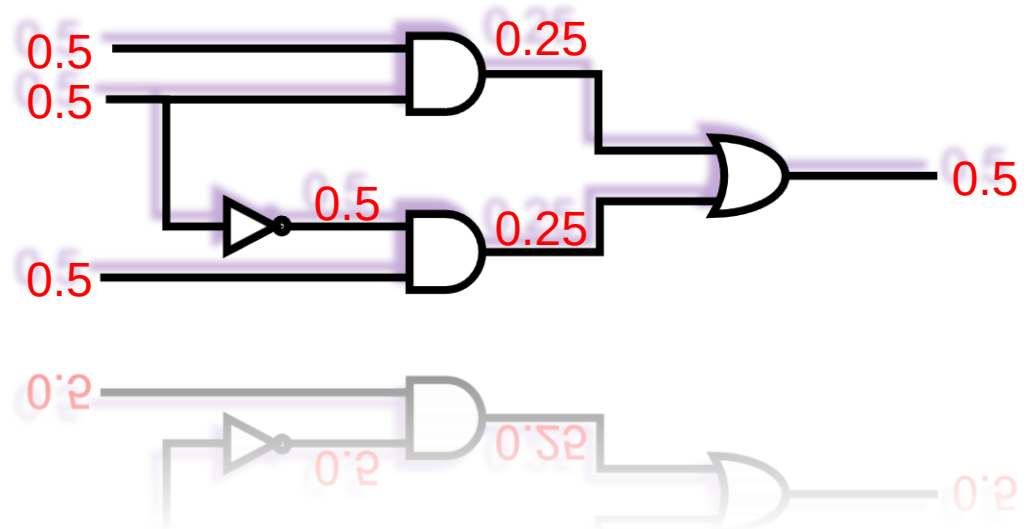
- What is *Testability Analysis*?
  - ♦ Calculate testability measures for a given circuit
- Why testability analysis?
  - ♦ 1. Help ATPG make smart decision
  - ♦ 2. Insert DFT circuits to improve controllability and observability
- When testability analysis?
  - ♦ In preprocess stage of ATPG or DFT insertion
- Requirements of testability analysis
  - ♦ Should run very fast
  - ♦ Sometimes , accuracy is not very important

# Categories of Testability Analysis

- 1. **Topology-based analysis**
  - ♦ Only analyzes structure of circuit. No test vectors are given.
  - ♦ Example: SCOAP
- 2. **Probability-based analysis**
  - ♦ Uses *signal probability* to estimate the testability
  - ♦ Example: COP
- 3. **High-level analysis**
  - ♦ Performs testability analysis before synthesis
  - ♦ Example: RTL testability analysis
- 4. **Simulation-base analysis (not in lecture)**
  - ♦ Apply input patterns,
  - ♦ Perform simulation and estimate testability

# Testability Measure

- Introduction
- SCOAP (1979)
  - ♦ Combinational
  - ♦ Sequential
- COP (1984)
- High-level testability measures



# SCOAP [Goldstein 1979]

- **Sandia Controllability Observability Analysis Program\***
- SCOAP computes 6 numbers for each node  $N$

\*Sandia is name of research lab.

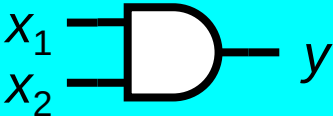
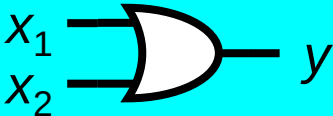
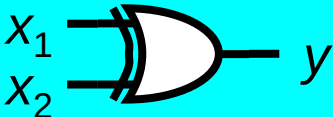
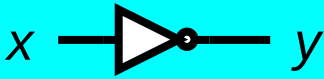
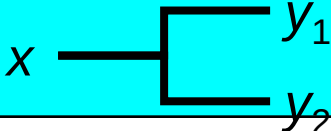
|               | 0-controllability | 1-controllability | Observability |
|---------------|-------------------|-------------------|---------------|
| Combinational | $CC^0(N)$         | $CC^1(N)$         | $CO(N)$       |
| Sequential    | $SC^0(N)$         | $SC^1(N)$         | $SO(N)$       |

# Combinational Controllability

- $CC^0(N)$ ,  $CC^1(N)$ 
  - ♦ minimum number of combinational PI assignments and logic levels required to control a 0 or a 1 on node  $N$
  - ♦ *Smaller* number, *easier* to control
- How to calculate CC?
  - ♦  $CC(PI) = 1$ .
  - ♦ From PI to PO. Add 1 to account for logic level
- Gate propagation rules:
  - ♦ If only one input controls gate output:
    - \*  $CC(\text{gate\_output}) = \min \{ CC(\text{gate\_input}) \} + 1$
  - ♦ If all inputs needed to set gate output:
    - \*  $CC(\text{gate\_output}) = \sum CC(\text{gate\_input}) + 1$
  - ♦  $CC(\text{branches}) = CC(\text{stem})$

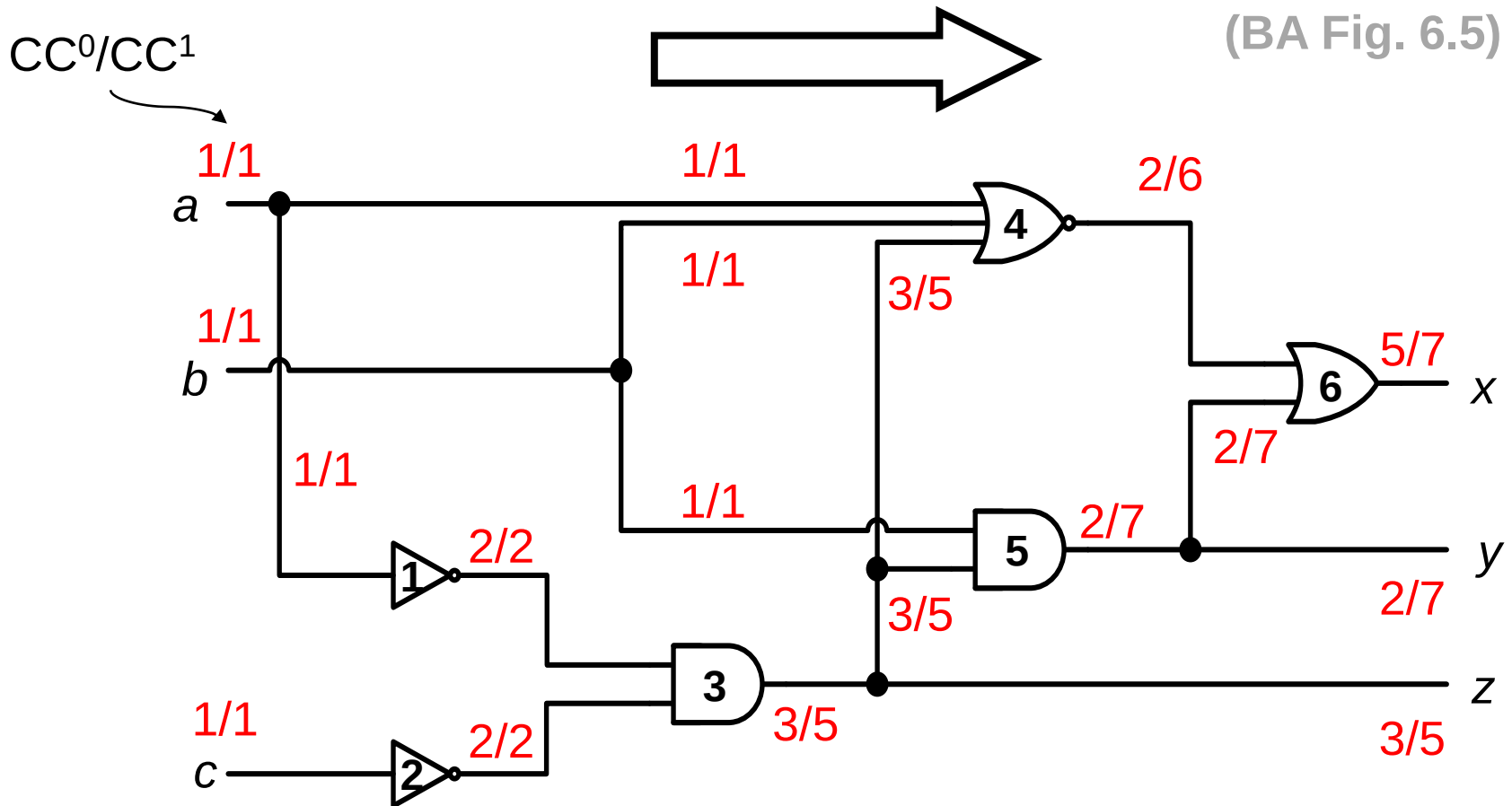
**Smaller CC is Better**

# CC<sup>0</sup>(N) & CC<sup>1</sup>(N)

|                                                                                     | CC <sup>0</sup> (y)                                      | CC <sup>1</sup> (y)                                      |
|-------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|
| Primary inputs                                                                      | 1                                                        | 1                                                        |
|    | $\min[CC^0(x_1), CC^0(x_2)] + 1$                         | $CC^1(x_1) + CC^1(x_2) + 1$                              |
|    | $CC^0(x_1) + CC^0(x_2) + 1$                              | $\min[CC^1(x_1), CC^1(x_2)] + 1$                         |
|   | $\min[CC^0(x_1) + CC^0(x_2), CC^1(x_1) + CC^1(x_2)] + 1$ | $\min[CC^0(x_1) + CC^1(x_2), CC^1(x_1) + CC^0(x_2)] + 1$ |
|  | $CC^1(x) + 1$                                            | $CC^0(x) + 1$                                            |
|  | $CC^0(y_1) = CC^0(y_2) = CC^0(x)$                        | $CC^1(y_1) = CC^1(y_2) = CC^1(x)$                        |

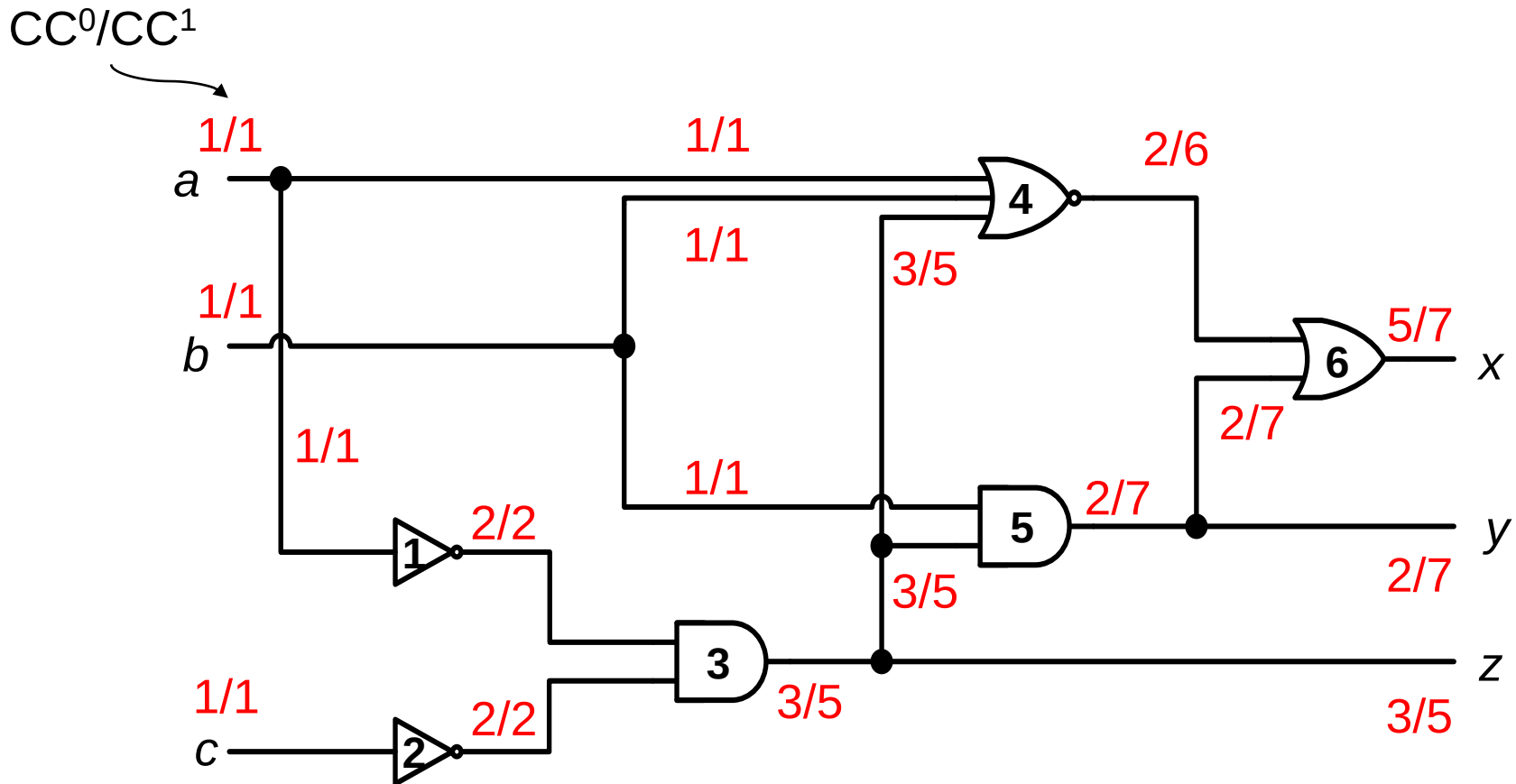
# An Example – Controllability

- Forward: From PI to PO



# Quiz

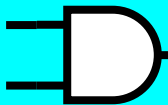
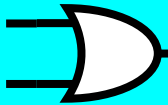
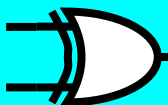
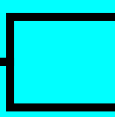
Q: how to control  $y$  to 0? How to control  $y$  to 1?



# Combinational Observability

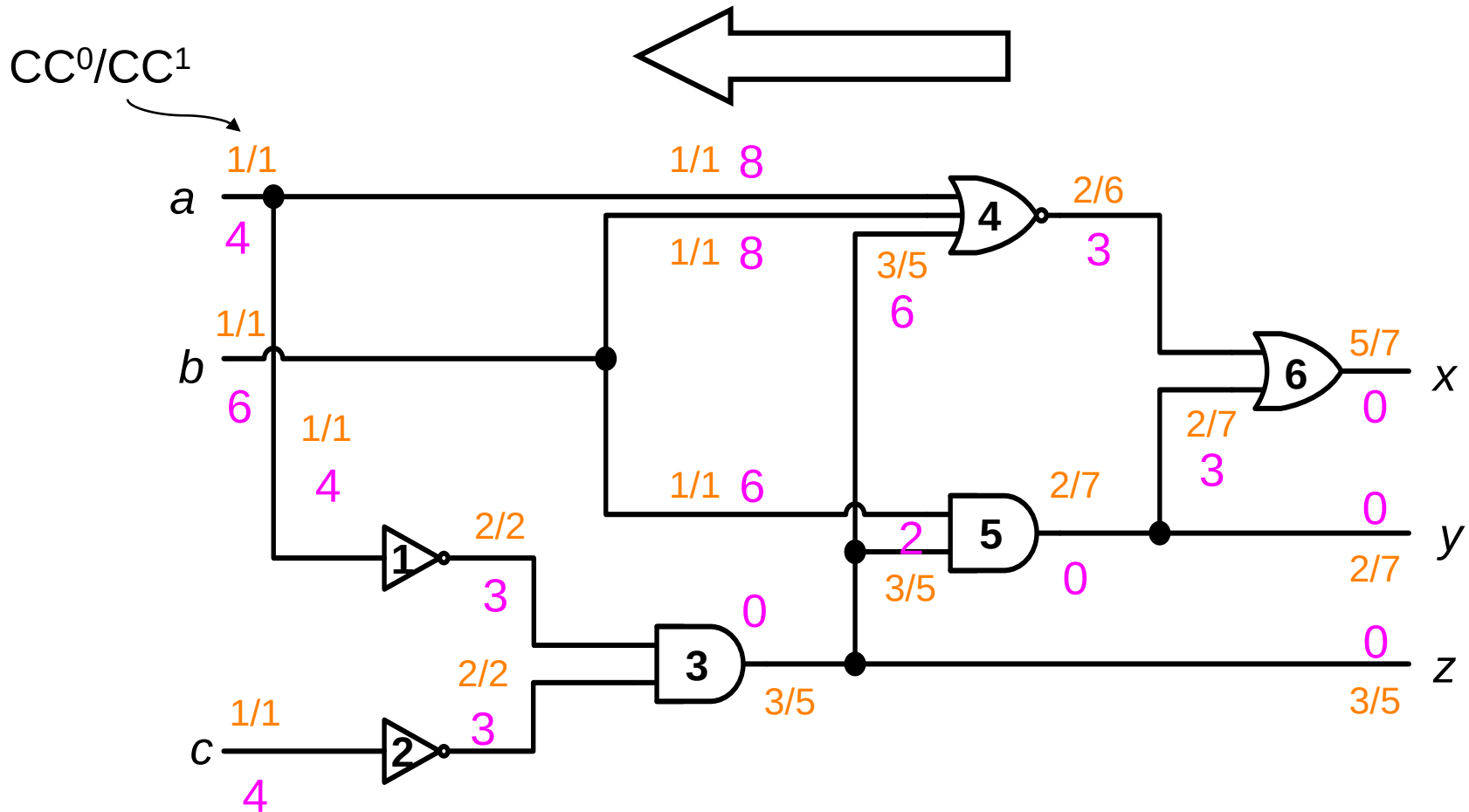
- **CO(N)**
  - ♦ minimum number of combinational PI assignments and logic levels required to propagate logical value on node  $N$  to PO
  - ♦ *Smaller* number, *easier* to observe
- **How to calculate CO(N)?**
  - ♦  $CO(PO)=0$
  - ♦ From POs to PIs, add 1 to account for logic level
- $CO(\text{gate\_input}) = \Sigma$ 
  - ♦ (1)  $CO(\text{gate\_output})$
  - ♦ (2) CC(setting all other inputs to non-controlling value)
  - ♦ (3) + 1 for logic level
- **How about fanout stem?** Assume they are *independent*
  - ♦  $CO(\text{stem}) = \min \{ CO(\text{branches}) \}$  *independent*  
 $CO(\text{stem}) = \max \{ CO(\text{branches}) \}$  *dependent*

# CO(N)

|                                                                                                          | CO( $x_1$ )                              |
|----------------------------------------------------------------------------------------------------------|------------------------------------------|
| Primary outputs                                                                                          | 0                                        |
| $x_1$<br>$x_2$  $y$     | $CO(y) + CC^1(x_2) + 1$                  |
| $x_1$<br>$x_2$  $y$     | $CO(y) + CC^0(x_2) + 1$                  |
| $x_1$<br>$x_2$  $y$     | $CO(y) + \min[CC^0(x_2), CC^1(x_2)] + 1$ |
| $x_1$  $y$            | $CO(y) + 1$                              |
| $x_1$  $y_1$<br>$y_2$ | $\min[CO(y_1), CO(y_2)]$                 |

# An Example – Observability

- Backward: From PO to PI

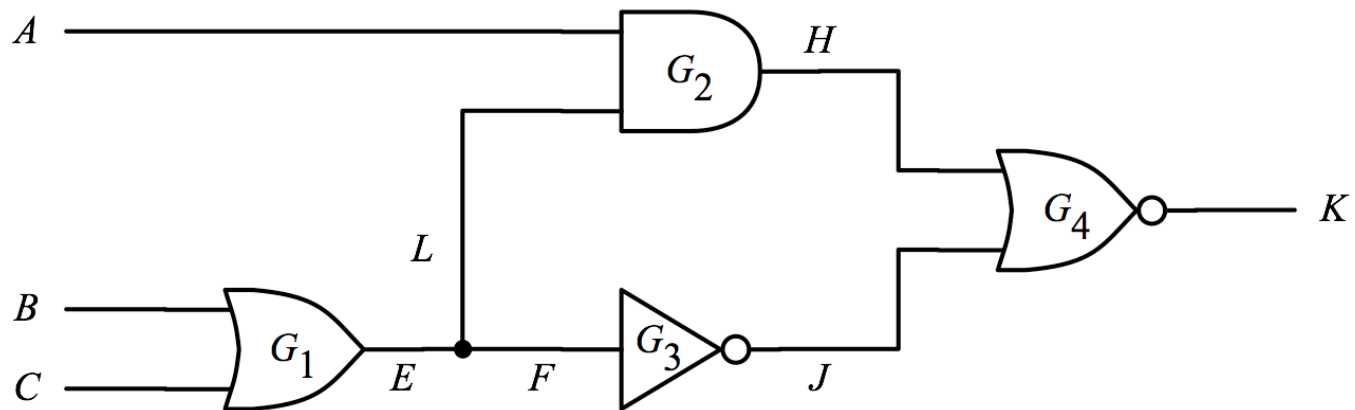


# Quiz

**Q: Please analyze combinational SCOAP.**

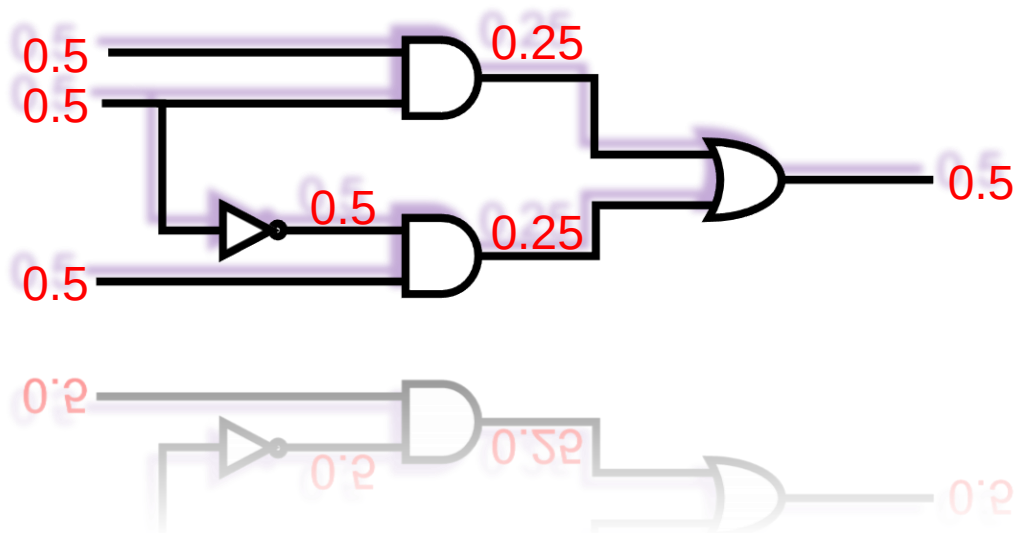
**A:**

|                 | A | B | C | E | F | L | H | J | K |
|-----------------|---|---|---|---|---|---|---|---|---|
| CC <sup>0</sup> |   |   |   |   |   |   |   |   |   |
| CC <sup>1</sup> |   |   |   |   |   |   |   |   |   |
| CO              |   |   |   |   |   |   |   |   |   |



# Summary

- Introduction
  - ◆ Metric to **measure degree of difficulty to test a circuit**
  - ◆ Helps **1. ATPG 2. DfT**
- SCOAP: Combinational
  - ◆  **$CC^0$ ,  $CC^1$ , CO**



# FFT

- **Q: Testability should be done very quickly. What is time complexity to calculate CC and CO?**

